

2026 AI Placement & Technical Interview Exam Portal

A Comprehensive Placement Assessment Engine Calibrated to 2026 Campus Hiring Standards

Repository:	https://github.com/likithyadav128-tech/ai-interview-exam-portal
Core Benchmark:	FACE Prep's 'The 2026 AI Career Roadmap for Indian Engineering Students'
Target Audience:	B.E. / B.Tech (CSE, IT, ECE, EEE, AIDS) & Recent Graduates (2025-2026)
Tech Stack:	React 18, Vite 6, Tailwind CSS, Lucide Icons, Python 3.11+, Streamlit 1.35+
Deployment Targets:	Streamlit Cloud, Vercel Edge, GitHub Pages, Netlify

1. Executive Summary & Problem Context

Between 2023 and 2026, campus placements across Indian engineering colleges underwent a monumental shift. Tier-1 IT services firms transitioned from mass baseline hiring (■3.5–4.5 LPA) to dedicated, high-paying AI tracks (such as **TCS Prime at ■7.0–11.5 LPA**, **HCLTech Elite at ■18–22 LPA**, **Infosys Power Programmer at ■6.5–9.5 LPA**, and **Wipro CoE**). Public statements from TCS confirmed that 60% of new joiners are now hired into AI-skilled tracks.

However, most undergraduate engineering students lack a dedicated, rigorous testing environment to benchmark their readiness across the 4 core layers of AI engineering (Python Foundations, ML Literacy, LLM Fluency, and Production System Integration). This project resolves that bottleneck by providing an end-to-end interactive examination portal, company-specific mock tests, an AI mock interview panel, an in-browser live code execution bench, and an automated CTC placement diagnostic engine.

2. System Architecture & Tech Stack

The application is architected as a modular Single-Page Application (SPA) with an inlined Python Streamlit runtime option for instant zero-configuration multi-platform cloud hosting.

Architectural Component	Underlying Technologies	Key Functional Purpose
Client UI Layer	React 18, Vite 6, Tailwind CSS	Ultra-responsive dark theme, glassmorphic HUD, interactive question palettes.
Examination Engine	Custom React Hook State Engine	Timed countdowns, flagged questions, single-choice evaluation, answer reviews.

AI Interview Room	Heuristic & Keyword Panel Evaluator	Grades 90s response pacing, metric presence, technical depth, and trade-offs.
In-Browser Code Runner	Python/JS Sandbox Engine	Live test execution for RAG chunkers, cosine similarity, and retry decorators.
Deployment Wrapper	Python Streamlit + Vercel SPA	Enables 1-click execution on Streamlit Community Cloud and native Vercel Edge.

3. Core Functional Modules

3.1 4-Layer Skill Stack Mock Exams

- **Layer 1 (Programming Fundamentals):** Python generators (lazy $O(1)$ space complexity), decorators, mutable default traps, Git conflict resolution, and virtual environment isolation.
- **Layer 2 (ML Literacy & Applied Math):** Linear/logistic regression, cross-validation, L1 (Lasso) vs L2 (Ridge) regularization, backpropagation chain rule intuition, and metric selection (F1, ROC-AUC).
- **Layer 3 (LLM Fluency & GenAI):** RAG architectures (chunking with overlap, vector DBs), prompt engineering (Chain-of-Thought), structured JSON outputs, LangGraph agents, and fine-tuning vs RAG trade-offs.
- **Layer 4 (System Integration):** FastAPI async request handling, Docker multi-stage builds, cloud deployment, regression eval suites, semantic token caching, and token cost modeling.

3.2 14 Tracked Company Round Simulators

Provides dedicated mock screening papers for the 14 companies tracked in 2026: **TCS Prime** (DSA + deployed AI project screen), **HCLTech Elite** (■18–22 LPA GenAI & agentic program), **Infosys Power Programmer** (LoRA parameter efficiency + graph algorithms), **D. E. Shaw India GAI Team** (Bayes' Theorem probability + transformer scaling $\frac{1}{\sqrt{d_k}}$), **Mu Sigma** (Decision analytics + muTalos case study), **Wipro CoE**, **Accenture GenAI Studio**, and **Tech Mahindra Project Indus**.

3.3 AI Technical Mock Interview Simulator

Rehearses the 5 universal placement questions every 2026 panel asks: (1) 90-second project walkthrough, (2) Fine-tuning vs RAG trade-offs, (3) LLM system evaluation suites, (4) Honest project limitations & engineering trade-offs, and (5) Company-specific pitch. Includes the *Business Outcome Translator Challenge* testing candidates on translating technical buzzwords into business outcomes (latency, users, cost).

3.4 In-Browser Live Coding Bench

Features an interactive code editor and automated test runner testing high-frequency placement coding problems: (1) Custom RAG text chunker with overlap, (2) Vector cosine similarity calculator from scratch, and (3) API retry decorator with exponential backoff.

3.5 5 Recruiter-Grade Project Blueprints & Resume Formulator

Provides 4-week execution plans for the 5 projects that change recruiter perception: Document Q&A; Bot, Resume Parser & Matcher, LangGraph Agent, Niche Fine-Tuned Model, and AI Placement Prep Tool. Features the dynamic *Rule 4.6 Resume Line Formulator* enforcing the rule: 'Every line ends with a number and real tooling'.

4. 2026 Fresher AI Salary & Diagnostic Matrix

Company Tier	Fresher CTC Band	Screening Criteria	Min Exam %
IT Services Baseline	■3.5 - 4.5 LPA	Aptitude + basic coding; AI is bonus	40%
IT Services AI-Tier (TCS Prime)	■6.5 - 11.0 LPA	Aptitude + 1 deployed AI project + coding	65%
Mid-size IT & GCCs	■5.0 - 9.0 LPA	1 to 2 AI projects + Python + basic SQL	60%
Product Companies Entry	■8.0 - 15.0 LPA	2+ deployed projects + system design basics	75%
IT Services Elite AI (HCLTech)	■18.0 - 22.0 LPA	Strong GenAI / agentic portfolio + extended screen	85%
Top Product & Quant (DE Shaw)	■35.0 - 50+ LPA	Premium DSA + advanced math / probability	90%

5. Deployment & Cloud Architecture

The repository is configured for multi-target zero-friction cloud deployment:

- Streamlit Community Cloud:** Execute via `app.py` with inlined React bundle on `share.streamlit.io`.
- Vercel Edge:** Instant deployment via `vercel.json` with single-page client-side rewrite rules.
- GitHub Pages:** Automated CI/CD workflow via `.github/workflows/deploy.yml` on push to main.
- Netlify:** Static publishing from `dist/` with `public/_redirects`.

6. Project Verification & Conclusion

- **Production Build:** Vite compilation succeeded (307 kB JS, 30 kB CSS gzipped to ~98 kB total bundle).
- **Source Synchronization:** All 25 source files and assets committed and pushed to GitHub main branch.
- **Educational Impact:** Directly serves undergraduate engineering students seeking high-bracket AI campus placements.

Status: Production Ready	Release Version: 1.0.0 (May 2026 Edition)
Repository: github.com/likithyadav128-tech/ai-interview-exam-portal	Format: IEEE Standard Project Report